Transistor

- Transistor was invented by **John Bardeen**, Walter Brattain, and William Shockley in 1947 (got nobel prize)
- One of the most important invention in the area of science and technology.
- A **transistor** is a three-terminal semiconductor device that controls the flow of current.
- Transistor word has come from the two words TRANSFER-RESISTOR, low input resistance to high output resistance and vice versa .

Types of Transistor

- Bipolar Junction Transistor (BJT)
  – NPN
  – PNP
- Field Effect Transistor (FET)
  – JFET(Junction Field Effect Transistor)
  – MOSFET(Metal oxides semiconductor Field Effect Transistor)

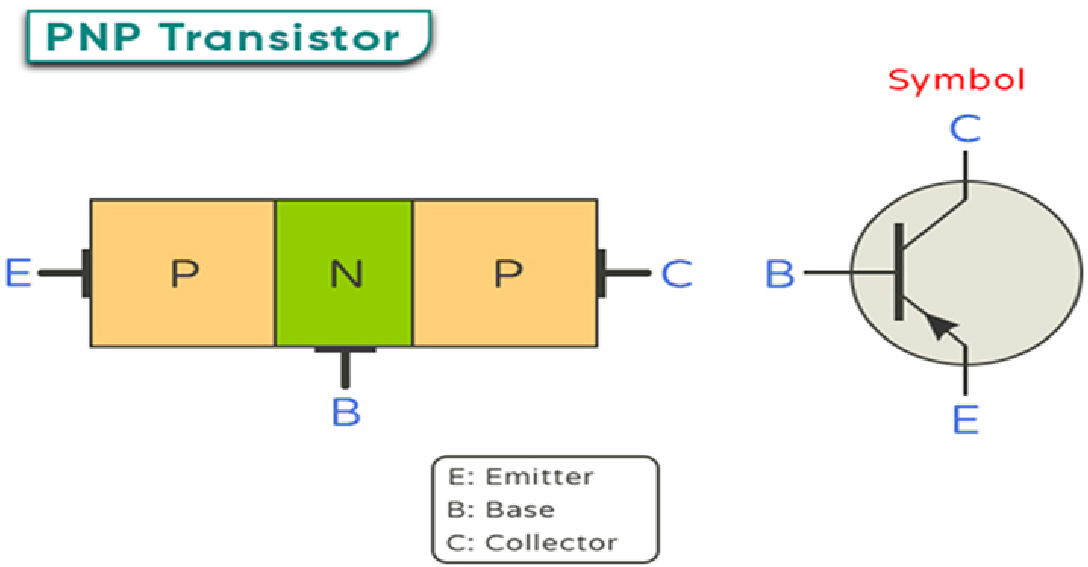
Constructional details of BJT

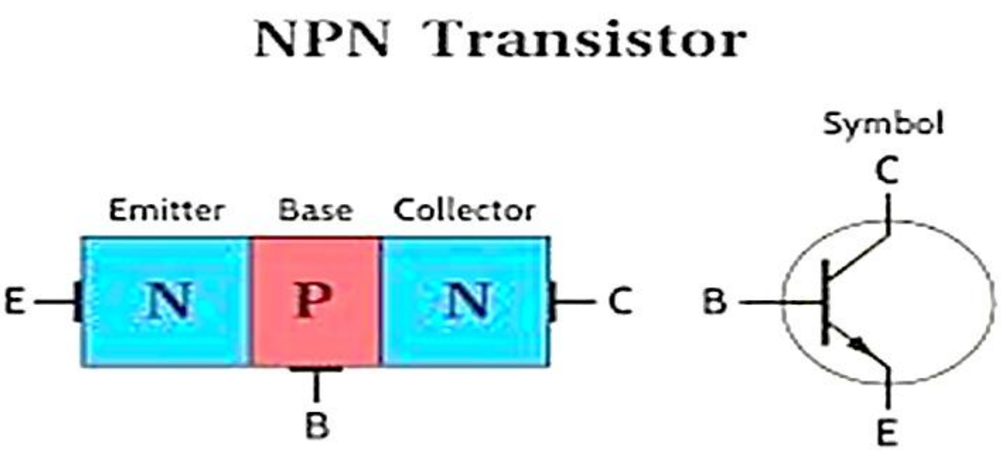
A transistor has **three Region and hence three terminals:**

1. **Emitter (E)** – emits charge carriers and it is heavily doped

2. **Base (B)** – controls the current, lightly doped and smaller size

3. **Collector (C)** – collects charge carriers either lightly or moderately doped, and larger in size as compared to emitter

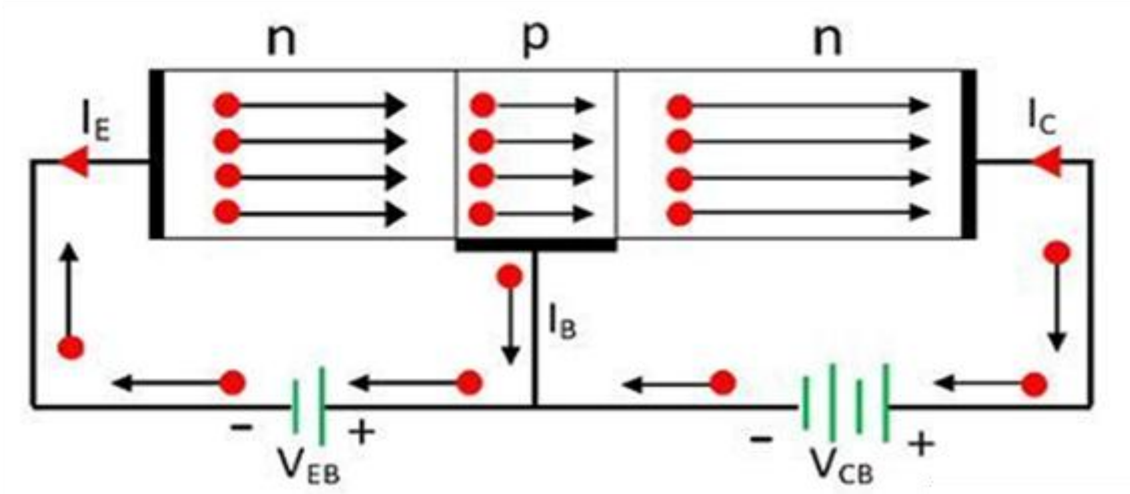




Important Points:

- Arrow is always on the emitter, and it indicates the direction of current for normal biasing of the transistors.
- Normally emitter base junction is forward biased and collector-based junction is reversed biased.

Working of the transistor



- Electrons are repelled from emitter and enter into the base region this causes emitter current which flows from the **emitter** into the **base**
- Since the base is thin A small number electrons recombine in the base, and the loss of electron is recovered by the supply of electrons from collector base bias voltage → **base current (I_B)**
- Remaining **most of the electrons are pass through it** and are attracted by the collector → **collector current (I_C)**
- Hence: $I_E = I_B + I_C$
- **For efficient working of transistor, collector current should be as large as possible component of emitter current.**

Transistor Configurations

Transistor is a three-terminal device. While using transistors for any applications having two input and two output terminals, one terminal is common to input as well as output.

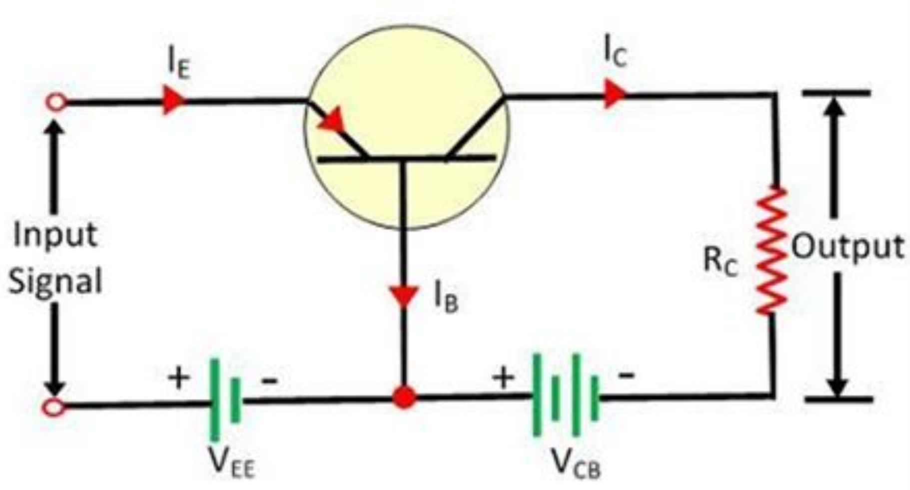
Transistors are used in the three configurations:

1. **Common Base Configuration**

2. **Common Emitter Configuration**

3. **Common Collector Configuration**

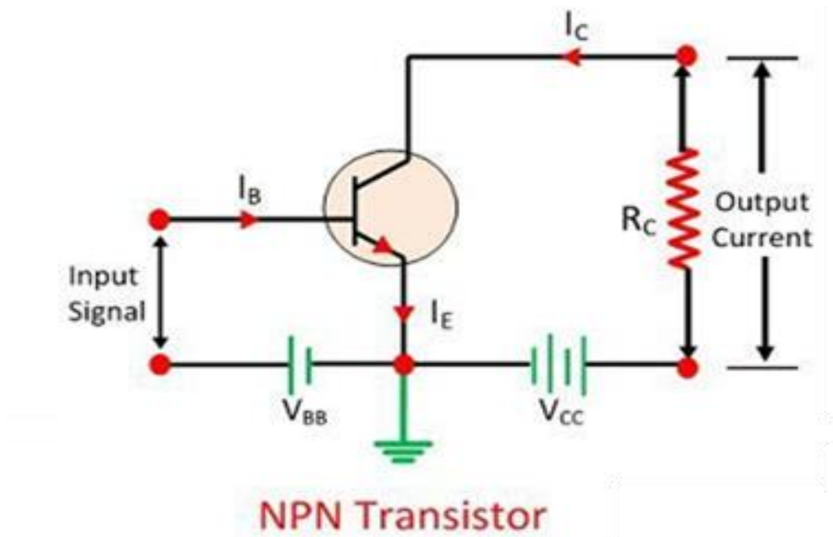
Common Base Configuration



Common Base Connection of PNP Transistor

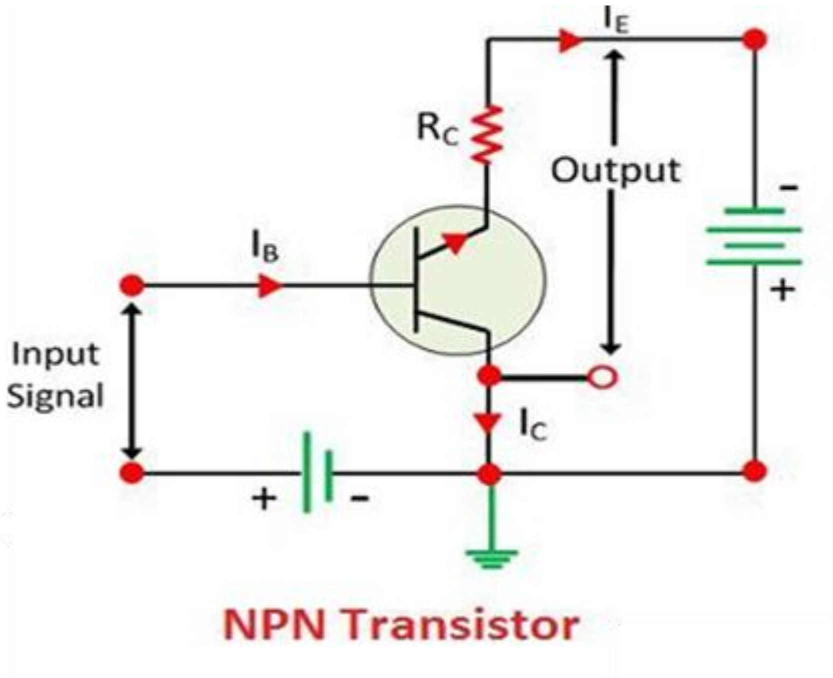
Input is given between emitter and base, and output is across collector and base. Base is common to input and output

Common Emitter Configuration



Input is given between base and emitter, and output is across collector and emitter. Emitter is common to input and output.

Common Collector Configuration



Input is given between base and collector, and output is across collector and emitter. Collector is common to input and output.

Current Gains for transistors

Alpha (α) of a Transistor:

Alpha (α) is the current gain or current amplification factor of a transistor in common base configuration.

Alpha (α) is defined as the ratio of collector current (I_C) to emitter current (I_E)

$$\alpha = \frac{I_C}{I_E}$$

Typically, close to one but always less than one

- **Beta (β)** is defined as the ratio of **collector current (I_C)** to **base current (I_B)** ,is current gain in common emitter configuration

$$\beta = \frac{I_C}{I_B}$$

- Typical values range from **20 to 300**
- A higher β means **greater amplification**
- α and β both are **dimensionless**

- **Relationship between α and β**

$$\beta = \frac{\alpha}{1-\alpha}$$

$$\alpha = \frac{\beta}{1+\beta}$$

The **Common collector (CC)** configuration provides the highest current gain among the three transistor set ups.

Current Gain (γ): It is defined as the ratio of the emitter current (I_E) to the base current (I_B) i.e $\boldsymbol{\gamma} = \frac{I_E}{I_B}$

Relationship with β: $\boldsymbol{\gamma = 1 + \beta}$

Since $\beta = \frac{I_C}{I_B}$ and $I_E = I_B + I_C$

the current gain in CC is slightly higher than in the common emitter configuration.

- Relation to Common Base Gain (α): $\gamma = \frac{1}{1 - \alpha}$
- The current gain is **very high**, typically ranging from 20 to several hundreds, depending on the transistor's β value

Characteristics of transistors

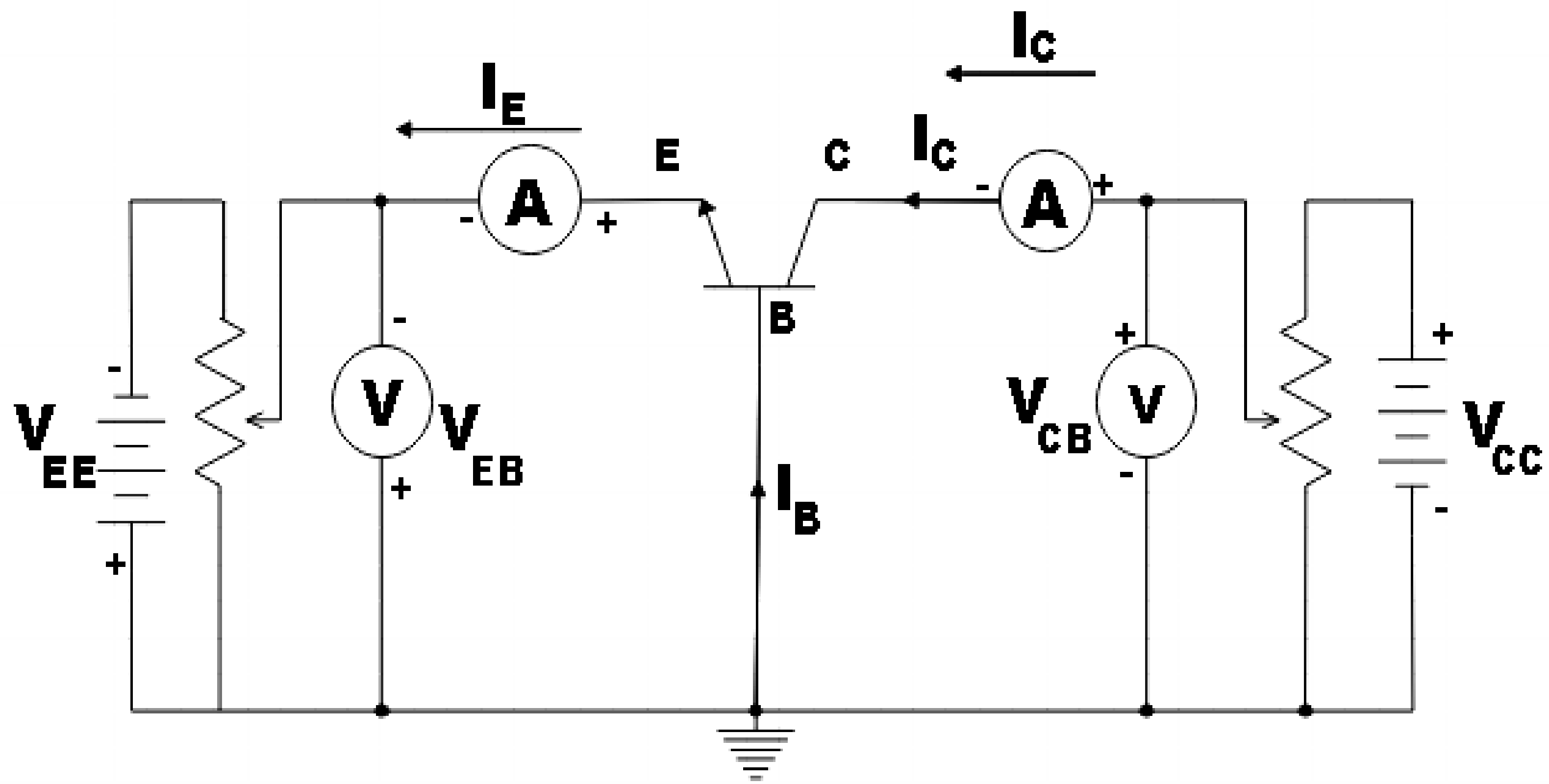
There are three types of transistors characteristics

1. Input Characteristics (Change in Input current due to change in input voltage at fixed output voltage)
2. Output Characteristics (Change in output current due to change in output voltage at fixed input current/voltage)

Transfer characteristics (The relationship between a device's **input** and **output**, usually as a graph plotting an output signal (voltage or current) against an input signal (voltage or current)

CB and CE configuration are most commonly used and are important.

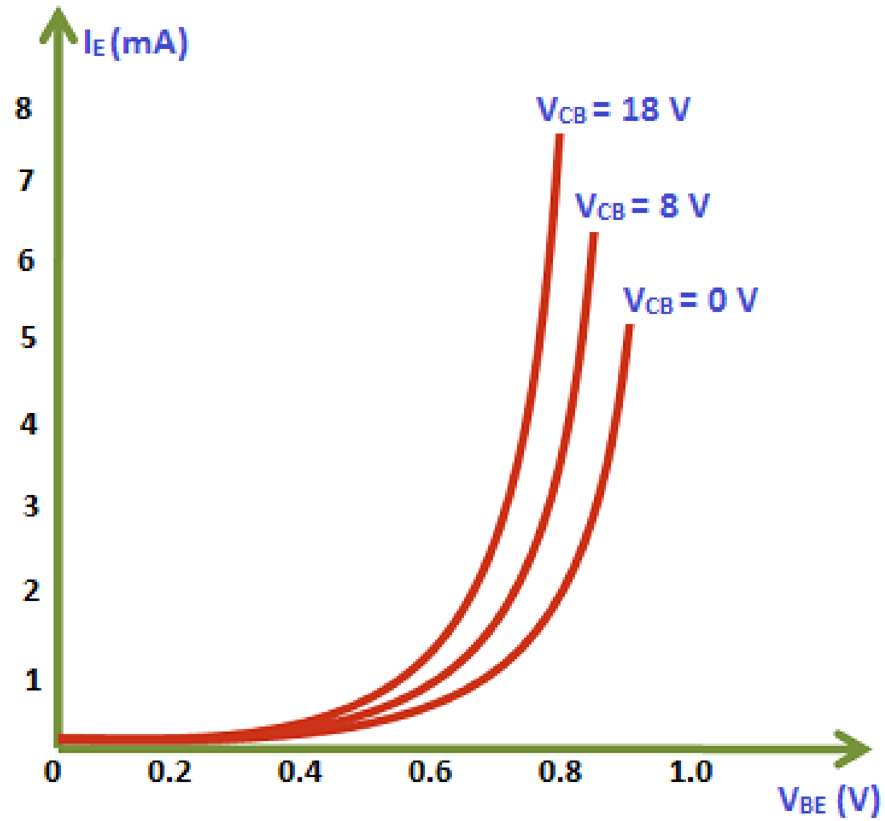
Common Base configuration



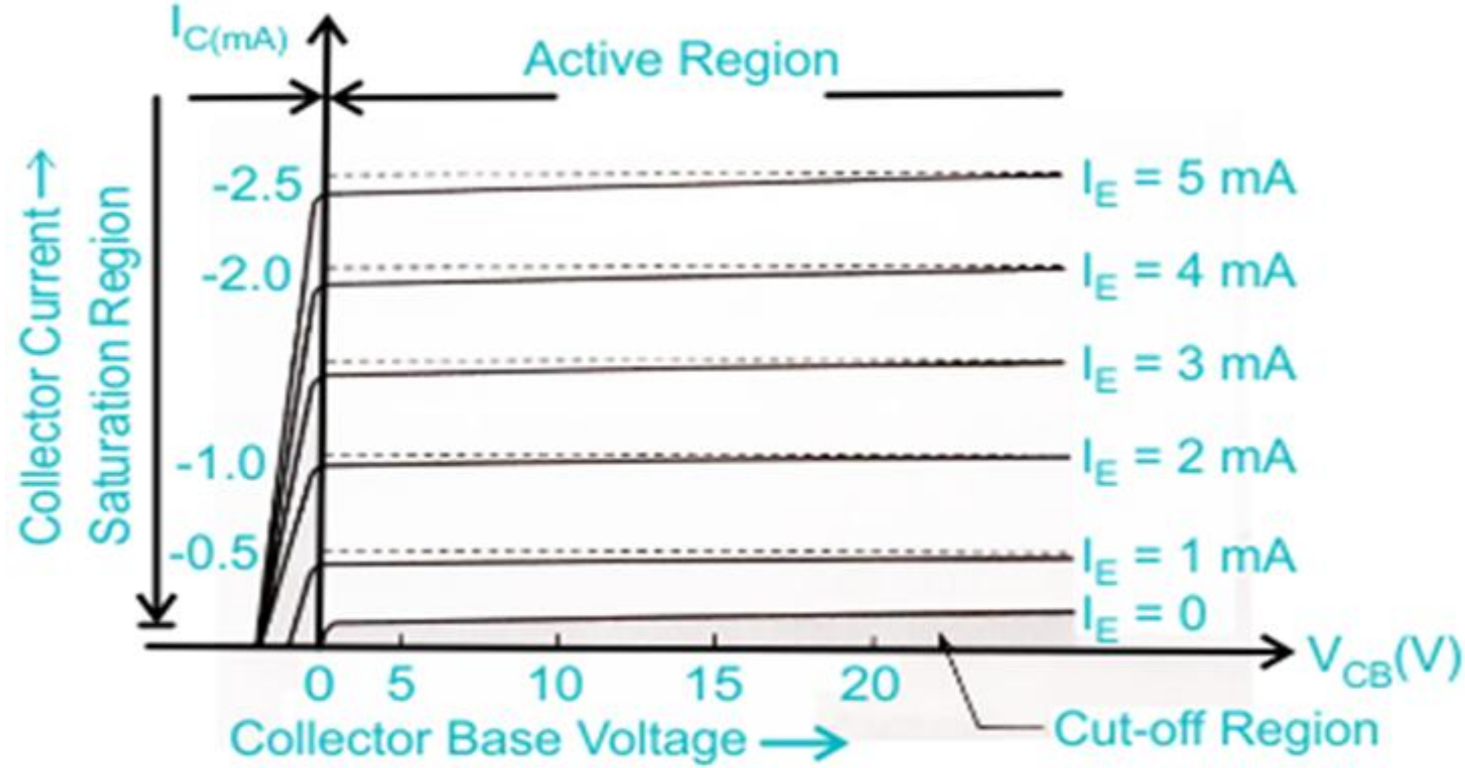
- It resembles a forward-biased diode curve (since the emitter-base junction is forward biased).

- A small increase in V_{BE} V causes a large increase in I_E.

- Input resistance Rin is low (typically 50–150 ohms).



I/p characteristics CB configuration



CB Output Characteristics

Common Emitter configuration

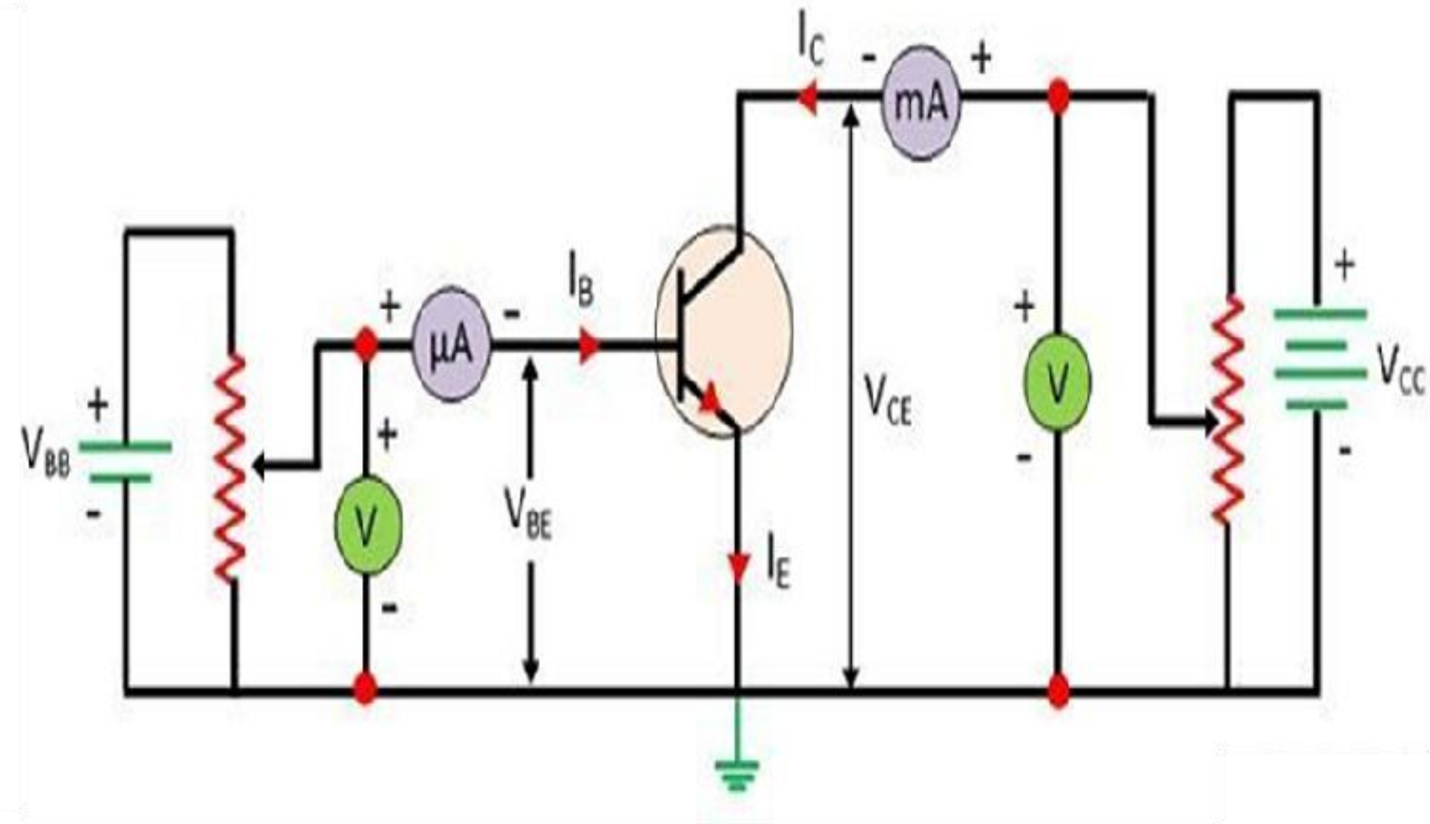
- For a given I_E, Ic remains nearly constant with increasing Vcb (indicating high output resistance).

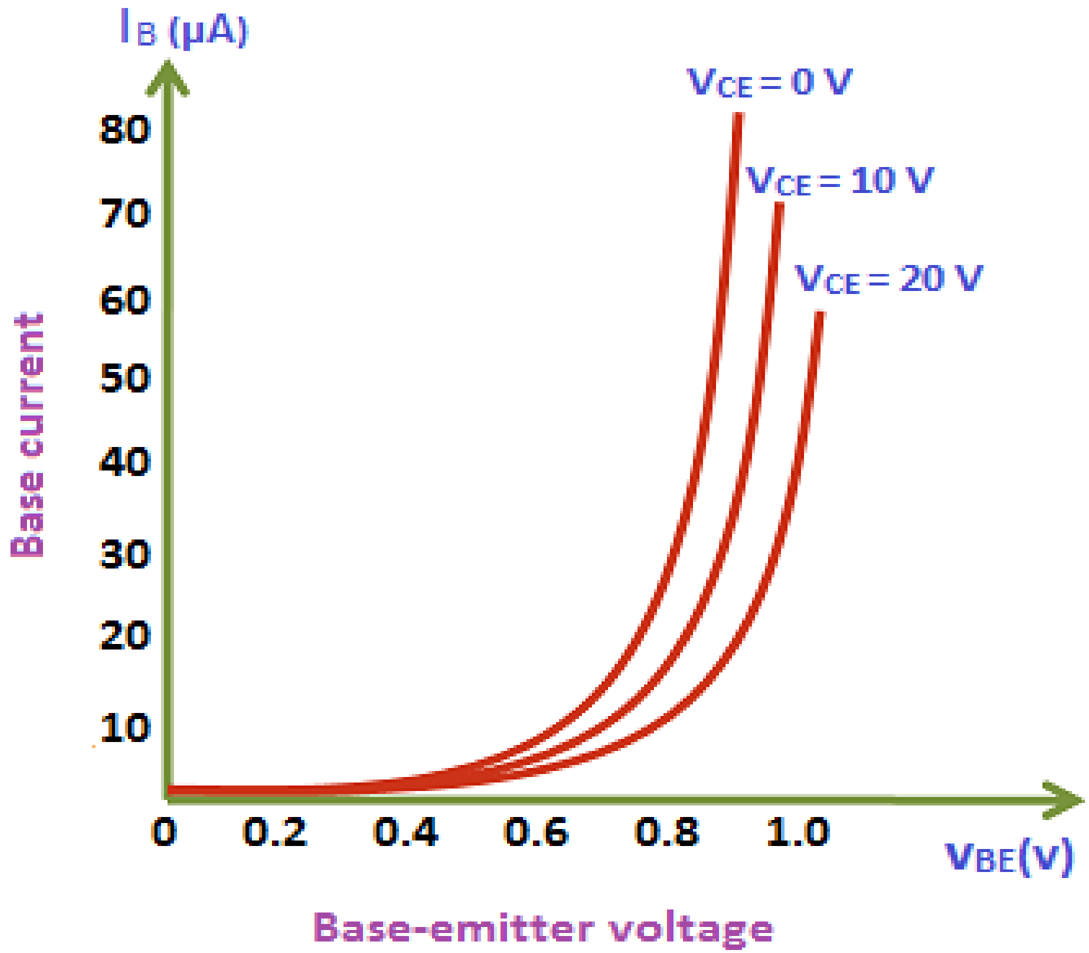
- Output resistance Rout is very high (in the range of several kilo-ohms).

- Slight slope in the curves indicates Early effect (collector current increases slightly with Vcb.

- The active region is where Veb is forward biased and Vcb is reverse biased.

Common Emitter configuration





I/P characteristics CE configuration

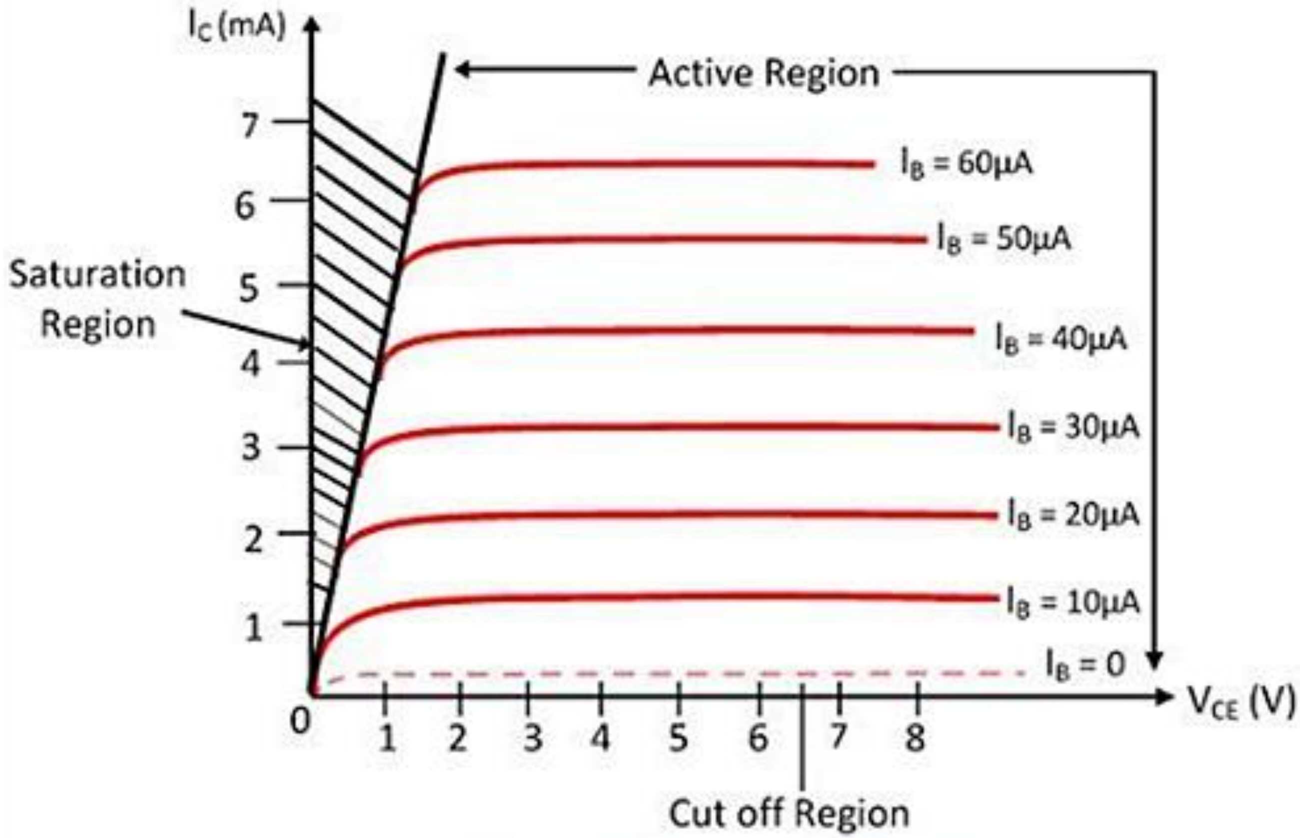
Characteristics Features

- **Diode-Like Curve**: The input characteristic curve resembles that of a forward-biased PN junction diode. This is because the base-emitter junction is forward-biased during normal operation
- **Knee (Threshold) Voltage**: There is negligible base current until the base-emitter voltage (V_{BE}) reaches a specific threshold, typically 0.7V for silicon and 0.3V for germanium transistors.
- **Exponential Current Increase**: Once V_{BE} exceeds the knee voltage, the base current (I_B) increases exponentially with relatively small increases in V_{BE}.
- **Moderate Input Resistance**: The dynamic input resistance (R_{in}) is defined as the ratio of change in V_{BE} to the change in I_B at constant V_{CE}
- **Value**: It is typically around **1 kΩ**.
- **Comparison**: This resistance is higher than that of the common-base configuration but lower than the common-collector configuration.

Early Effect

- **Effect of V_{CE} (Early Effect)**: As the collector-emitter voltage (V_{CE}) increases, the input characteristic curves shift slightly to the right. Increasing V_{CE} expands the collector-base depletion region, effectively narrowing the base region (base-width modulation or the **Early Effect**).

- This causes a slight decrease in base current for a given V_{BE} at higher V_{CE} levels.



Output Characteristic Curve

Common Emitter configuration O/P characteristics

- The output characteristic curve is divided into three distinct operational regions:
- **Active Region:**
  - **Condition:** Base-Emitter junction is forward-biased and Collector-Base junction is reverse-biased.
  - **Behavior:** The curves are nearly horizontal, meaning I_C is relatively constant and mainly **depends on I_B rather than V_{CE}**
  - **Application:** The transistor functions as an **amplifier** in this region.
- **Saturation Region:**
  - **Condition:** Both the Base-Emitter and Collector-Base junctions are forward-biased.
  - **Behavior:** I_C increases rapidly with a small increase in V_{CE} until it reaches a saturation point.
  - **Application:** The transistor acts as a **closed switch**.

- **Cut-off Region:**
  - **Condition:** Both junctions are reverse-biased (I_B).
  - **Behavior:** Only a very small leakage current (I_{CE}) flows, and I_C is nearly zero.
  - **Application:** The transistor acts as an **open switch**.

- **Key Parameters & Features**
- **Output Resistance (r_o):** This is the ratio of change in V_{CE} to the change in I_C at a constant I_BIn the active region, the output resistance is **high** (typically several kilo-ohms).
- **Current Gain (β** is the ratio of I_C to I_B (Explained earlier).
- **Early Effect:** A slight slope in the curves in the active region indicates that I_C increases marginally as V_{CE} increases due to the narrowing of the effective base width.

- **Phase Inversion:** This configuration is unique for producing a **180° phase shift** between the input and output signals.

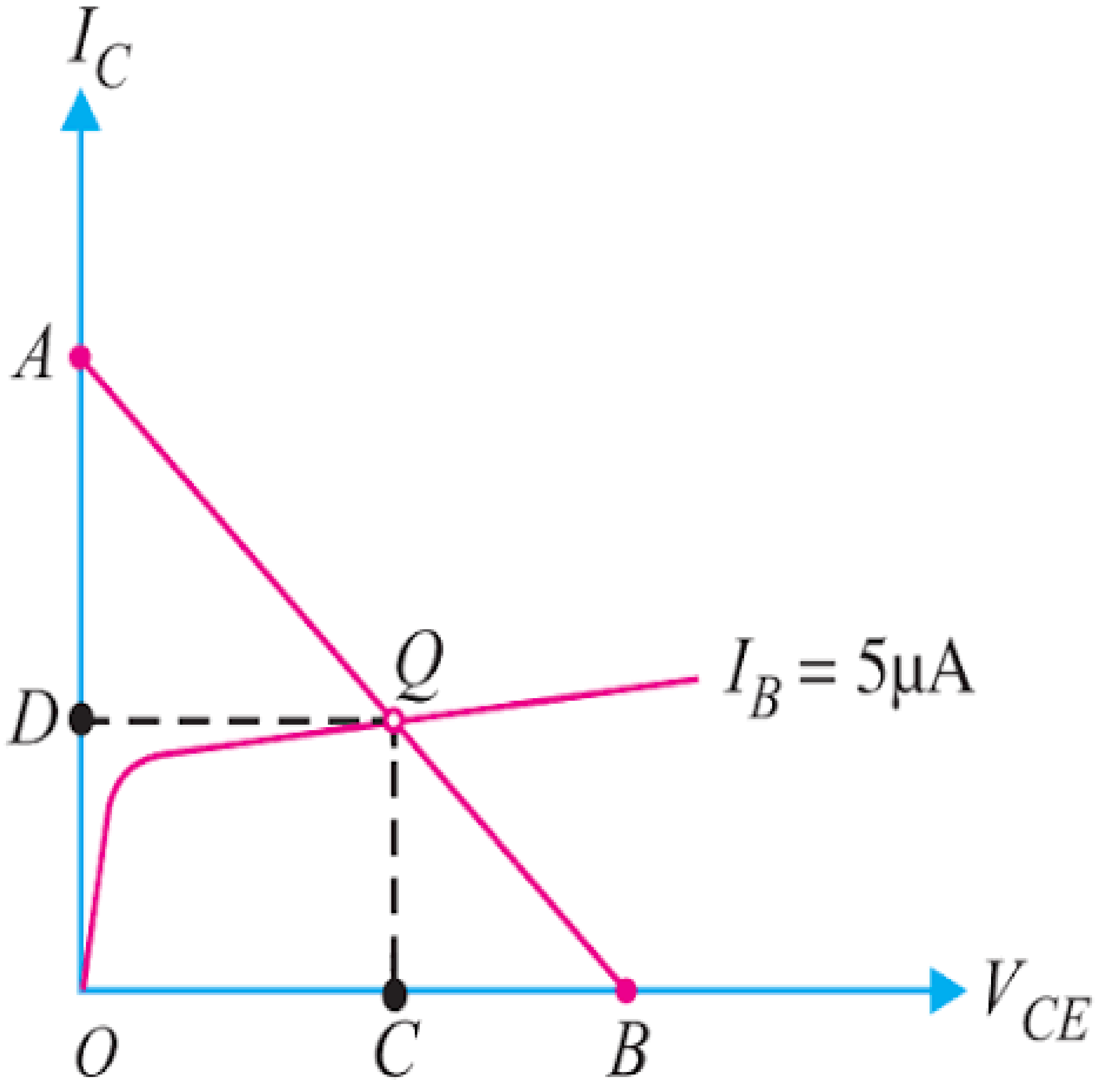
- **Gain Characteristics:** It provides high power gain because it produces significant current and voltage gains simultaneously.

Q Point

- **Q-Point:** Quiescent point or Q-point it is also called as the operating point of a device or bias point, is the steady-state DC voltage or current at a specified terminal of an active device such as a transistor with no input signal applied.
- The term ‘bias point’ is used because, sometimes, an electronic device like a transistor must be biased with a steady DC input so that it can operate correctly and without saturation or cutoff when performing its role as a processor of a time-varying AC signal (Amplifier, the most common application of the transistor).
- For example, a transistor could be biased to set its Q-point output to, say, 5 V to sit halfway between 0 and 10 V supply rails.
- An AC input then superimposed onto the DC level at the transistor input will generate a corresponding, amplified waveform at the output. Biasing provides room for this waveform either side of the Q-point without clipping or saturation.

- The position of the Q-point depends on the applications of the transistor. If the transistor is used as a switch, then for open switch the Q-point is in the cut-off region, and for the close switch, the Q-point is in the saturation region. The Q-point lies in the middle of the line for the transistor which operates as an amplifier.

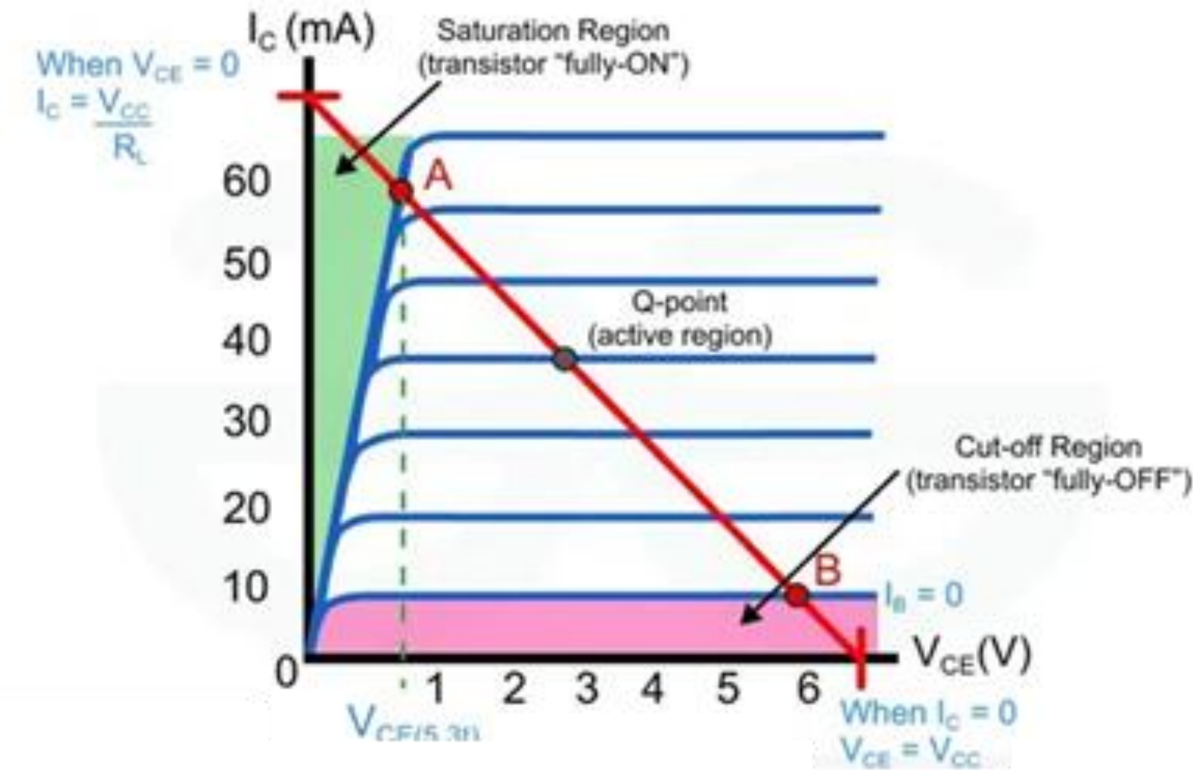
- In saturation region, both the collector base region and the emitter-base region are in forward biased and heavy current flow through the junction. And the region in which both the junctions of the transistor are in reversed biased is called the cut-off region.



I_C
A
Q
$I_B = 5\mu A$
D
V_{CE}
O
C
B

Transistor as a Switch

Transistor as a switch is the simplest application of the device. A transistor can be used for switching operations or opening or closing of the circuit.

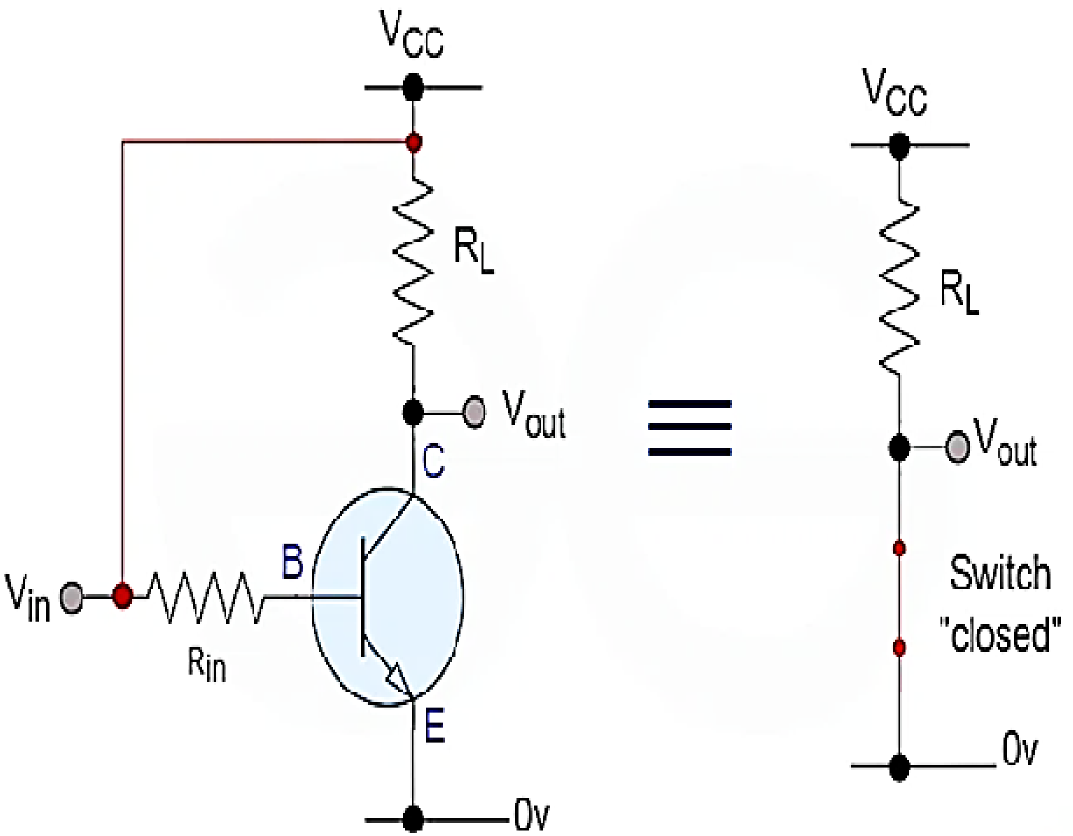


Closed switch

- **Saturation Region**

In this the transistor will be biased so the maximum amount of the current is applied, which results in the maximum collector current results in the minimum collector emitter voltage drop which cause in depletion layer as a small as possible and maximum current flowing through the transistor, so the transistor is switched "Fully ON".

- The base and input are connected to Vcc.
- Base-Emitter voltage Vbe>0.7V.
- Base-Emitter junction is forward biased.
- Base-collector junction is forward biased.
- Transistor is "fully-On"



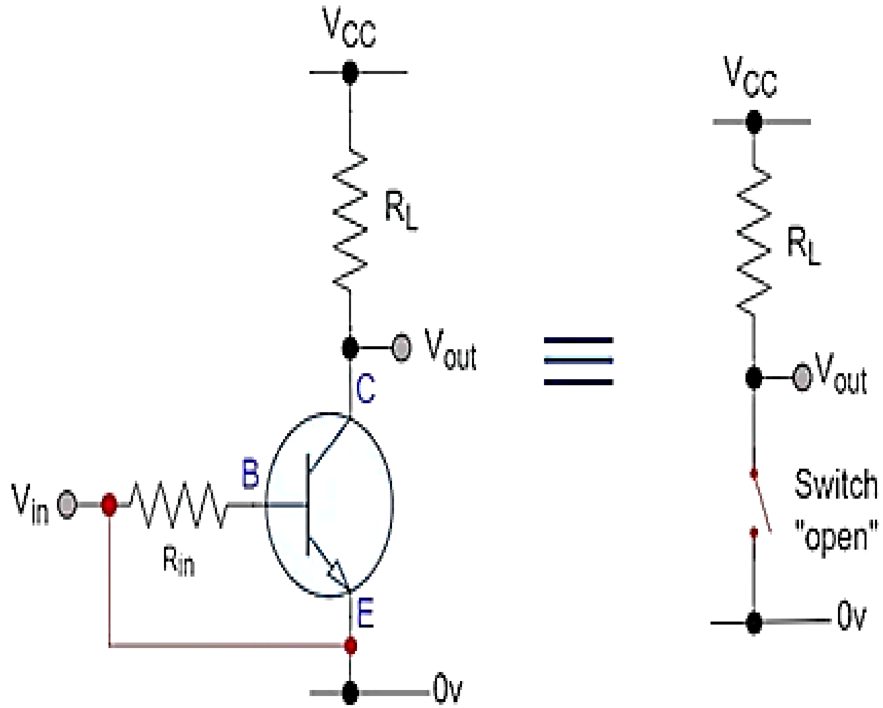
- Max Collector current flows (Ic=Vcc/Rl)
- Vce=0 (ideal saturation)
- Vout = Vce="0".
- Transistor operates as a closed switch.

Open switch

- **Cut-off Region**
- In the cut-off region of the transistor, it operates under conditions of zero input base current (Ib), zero output collector current (Ic), and maximum collector voltage (Vce). This configuration creates a significant depletion layer, preventing the flow of current through the device and effectively switching it off.

- The input and base are grounded.
- Base emitter voltage Vbe< 0.7V
- Base Emitter junctions is reversed.
- Base- collector junction us reversed biased.
- Transistor is "fully-Off" (cut off region)
- No collector current flows(Ic=0)

- Vout = Vce= Vcc= 1



- Transistor operates as open switch.
  The cut off region of off mod when using a bipolar transistor as a switch